SMART BUS SYSTEM FOR REAL TIME PASSENGER MANAGEMENT

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Meet our team

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PP2 Comments and Actions Taken

PP2 Comments Received	Action Taken
Common comment for all 4 members :- ▪ Collect more data by fixing the system to a real-world bus environment	✓ Expanded the data collection plan by meeting bus drivers and conductors and gathering real data through questionnaires. ✓ Prepared real-world testing with buses and collected practical data.
Abeysekara W.R.G.M comment :- ▪ Arrange all components and present a proper demo	✓ Integrated all four modules into one complete system ✓ Improved end-to-end demonstration flow

Research Problem

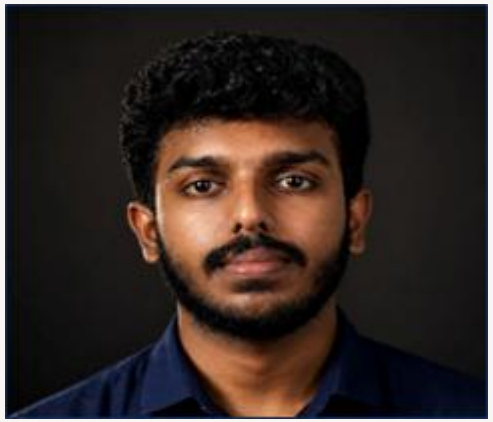
- Existing bus tracking systems provide **limited real-time information** and do not integrate:
  - Passenger crowd monitoring
  - Accurate ETA prediction
  - Passenger–driver communication
- Passengers face:
  - Unpredictable arrival times
  - Severe overcrowding
  - Lack of real-time information
- Transport authorities lack data-driven decision systems.

Design Excellence of the Overall Solution

- Real-time GPS Bus Tracking
- Accurate ETA Prediction
- Passenger Crowd Monitoring
- Mobile Passenger Communication System
- Fleet Route Optimization
- Vision-Based Occupancy Detection
- User-Friendly Mobile Application

IT22131560 | WEERAKOON W.M.D.P

IoT Based Bus Tracking System And ETA Prediction



Problem

- Passengers often face uncertainty in bus arrival times.
- If we can predict accurate arrival time of bus, it will save so much time of public transport users.

Solution

- Collects live GPS data from buses
- Tracks current location in real time
- Predicts estimated arrival time (ETA)
- Sends live updates to passengers

IT22230492 | PATHIRAJA H.P.M.O.N.

Central Fleet Management and Route Optimization Engine

Problem

- Passengers experience unexpected delays due to inaccurate ETA for their destination
- Lack of real-time predictions for buses reach passenger's destination
- Lack of transparency in ticket pricing.
- No real-time system to report driver behavior, reducing passenger safety.

Solution

- Mobile UI to show prediction results after entering “From” and “To” destinations
- Using real-time IoT data bus and Google Maps API.
- Pre-journey ticket fare estimation based on approved pricing rules.
- Real-time passenger complaint system with driver alerts.

IT22286246 | CHANDRASEKARA C.M.P.V

Mobile-Based Passenger Interaction System

Problem

- Passengers face uncertainty about bus arrival time
- Overcrowding causes discomfort, delays, and safety concerns.
- Lack of real-time communication between passengers and drivers.
- No crowd forecast for better journey planning.

Solution

- Mobile app for passengers
- Mobile-based ML crowd prediction system
- Simple output messages for decision making
- Passenger-to-driver messaging feature

IT22271150 | Abeysekara W.R.G.M

Vision based passenger counting and Occupancy

Problem

- Users cannot know how many seats available before boarding.
- Overcrowding causes discomfort, delays, and safety concerns.
- Lack of real-time seat occupancy and crowd checking.
- Authorities make decisions based on guesses instead of real data.

Solution

- Counts passengers entering/leaving bus
- Detects overcrowding levels
- Sends occupancy status to system

Results & Testimonials

Results

- Real-time bus tracking successfully implemented
- Accurate ETA prediction achieved
- Occupancy monitoring working effectively
- Mobile application fully functional
- Successful integration of all modules

User Feedback

- Very useful to know bus arrival times.
- Helps avoid crowded buses.
- Easy and simple mobile interface.

Benchmark Results

Feature	Existing Systems	Our System
Real-time Tracking	Limited	Yes
Occupancy Detection	No	Ye
ETA Prediction	Basic	Advanced
Passenger Communication	No	Yes

Limitations & Future Improvements

Current Limitations

- Requires stable internet connection
- GPS signal issues in some rural areas
- Camera performance depends on lighting conditions

Future Improvements

- AI traffic prediction integration
- Multi-language mobile app
- Expansion to railway transport systems
- Cashless payments
- Online ticket booking system

Thank You!